

DATA SCIENCE CAPSTONE PROJECT

Hidden Gems Hotels Recommender

A content-based hotel recommendation system for surfacing high-quality,
underappreciated European stays

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EXECUTIVE SUMMARY

Executive Summary

The Hidden Gems Hotels Recommender is a content-based hotel recommendation system designed to surface high-quality but relatively underappreciated hotels in major European cities. Instead of ranking hotels only by popularity or review volume, the system combines review ratings, aspect-based sentiment analysis, quiet and crowded review signals, geographic features, and user preference matching.

For notebook development and reporting, the analysis uses a 100,000-review sample to support efficient local experimentation. The Streamlit application supports 100k reviews, 200k reviews, and full-dataset processing modes.

Project Goal

Identify strong hotel options that may be overlooked by traditional popularity-driven hotel rankings.

- The recommender separates baseline hotel quality from user-specific preference adjustments.
- Paris showed the highest average top-hotel hidden gem score among the evaluated cities in the reporting sample.
- Preference settings meaningfully changed results: 7 of the top 10 Paris recommendations changed between quiet and lively traveler scenarios.
- The completed Streamlit app provides city selection, preference controls, ranking tables, map visualization, and explainability metrics.

BUSINESS PROBLEM

Business Problem

Hotel discovery platforms often favor highly reviewed, heavily marketed, or chain properties. This can make smaller or less-publicized hotels difficult to discover, even when guests consistently rate them highly.

This project asks: which hotels provide excellent guest experiences while remaining relatively overlooked compared with competitors in the same city? The recommendation system is designed to act as a discovery layer that helps travelers find high-quality hotels that may not appear at the top of popularity-based rankings.

DATASET

Dataset

The project uses the Kaggle 515K Hotel Reviews Data in Europe dataset. It includes hotel names, reviewer scores, positive and negative review text, hotel coordinates, and total review counts for hotels in Amsterdam, Barcelona, London, Milan, Paris, and Vienna.

Reporting Scope

Notebook figures are based on a 100,000-review sample. The application provides selectable processing modes for 100k reviews, 200k reviews, and the full dataset.

METHODOLOGY

Methodology

Data Preparation

Reviews were cleaned and aggregated from the review level to the hotel level. The notebook workflow uses Province_Name renamed to Metro_Area as the city field rather than parsing Hotel_Address. Hotel-level features were then normalized where appropriate for scoring and within-city comparison.

Aspect-Based Sentiment Analysis

Instead of treating each review as a single sentiment value, the notebook applies aspect-based sentiment analysis. Review sentences are split, matched against aspect-specific keyword sets, scored using TextBlob, and aggregated to the hotel level.

- Aspects: noise, cleanliness, staff, location, and value.
- Generated features: noise_sentiment, cleanliness_sentiment, staff_sentiment, location_sentiment, and value_sentiment.

Quiet vs. Crowded NLP Features

Additional NLP features capture whether review language describes hotels as quiet, calm, relaxing, noisy, busy, or crowded. These features are especially important because the application supports traveler preference controls for quiet/calm and crowd/noise tolerance.

Geographic Features

Hotel coordinates were compared with fixed city-center coordinates using the Haversine formula. Distance was then transformed into several relative features so that users can prefer central, balanced, or farther-from-center hotel locations.

- distance_from_center_km
- distance_percentile_within_city
- distance_sweet_spot
- centrality_score

Hidden Gem Scoring

The baseline hidden-gem score combines hotel quality and review-derived features while applying a small penalty for high review volume. This creates a ranking that rewards strong guest experience while reducing overreliance on popularity.

Base Score

The baseline score rewards rating, cleanliness, staff, location, value, and noise sentiment while applying small penalties for crowded-language signals and log review count.

Personalized Recommendation Engine

The final recommender intentionally separates `hidden_score_base` from user preference matching. Preferences adjust rankings using calm quality, crowd match, and distance match scores. This keeps the baseline quality model interpretable while still allowing traveler-specific personalization.

- `quiet_match_score`
- `calm_quality_score`
- `crowd_match_score`
- `distance_match_score`
- `final_score`
- `city_final_percentile`

RESULTS

Results

The results section focuses on a small number of figures that support the project data story: hidden gem scores are measurable, popularity is not the same as quality, city-level opportunities differ, user preferences change rankings, and the recommendation logic has been implemented in an interactive application.

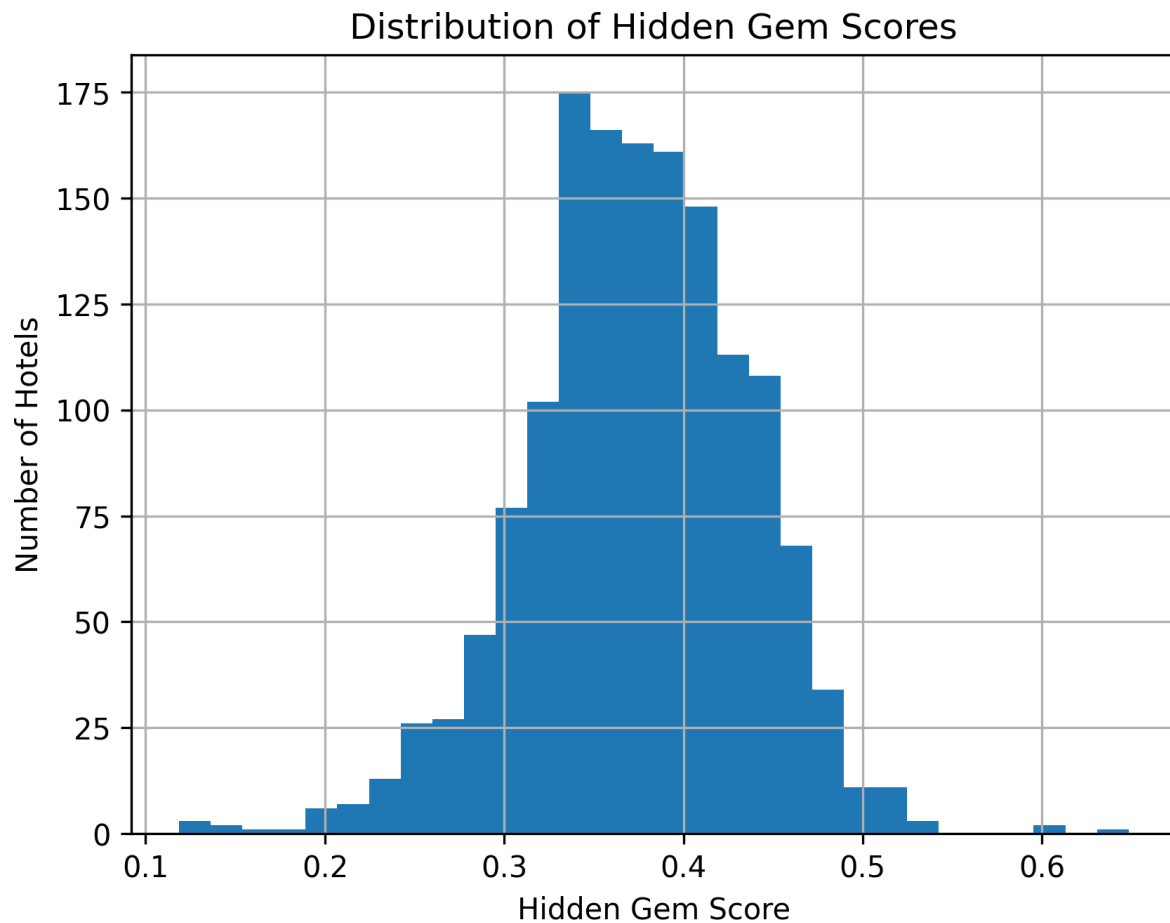


Figure 1. Distribution of `hidden_score_base` values in the 100k reporting sample.

Takeaway

Most hotels fall within a middle hidden-gem score range, while a smaller group separates as stronger hidden-gem candidates.

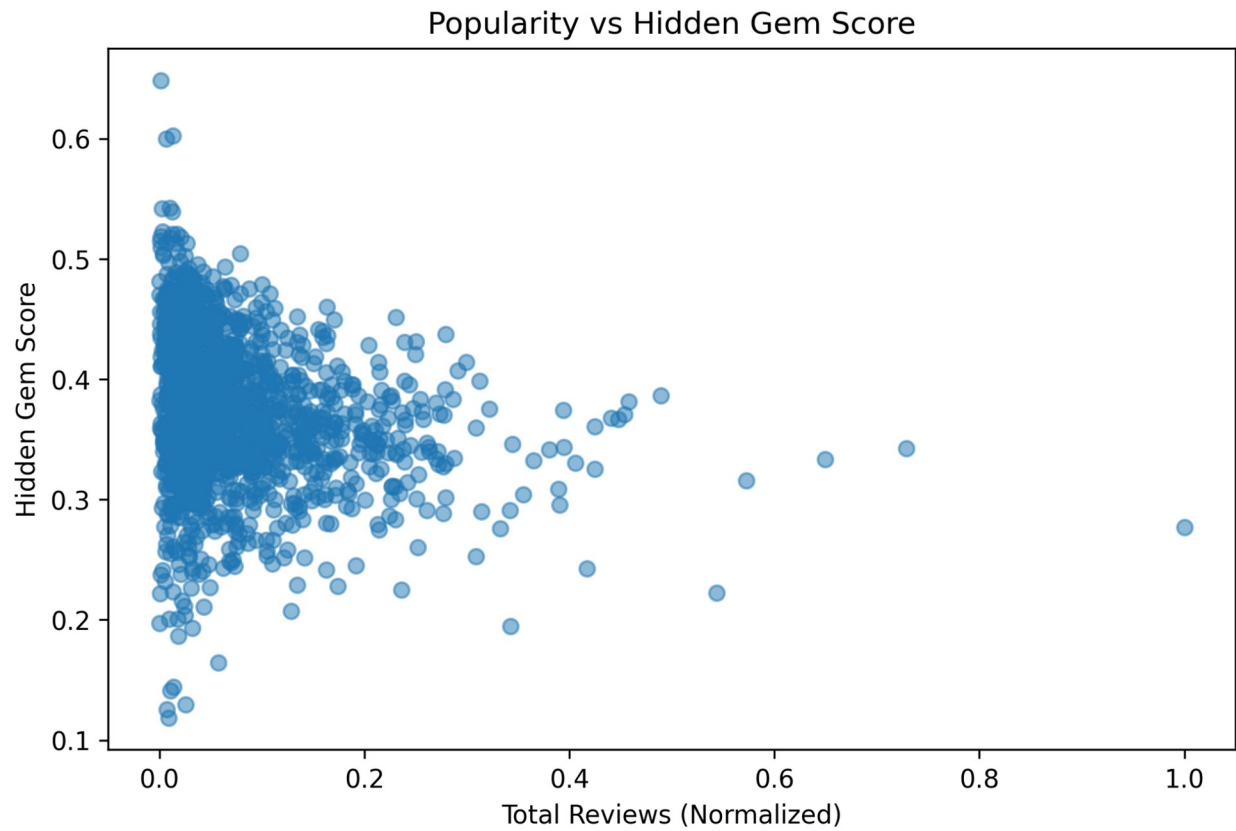


Figure 2. Popularity vs. hidden gem score. Total reviews are normalized.

Takeaway

Review volume alone does not explain hidden-gem score; some lower-popularity hotels still show strong quality signals.

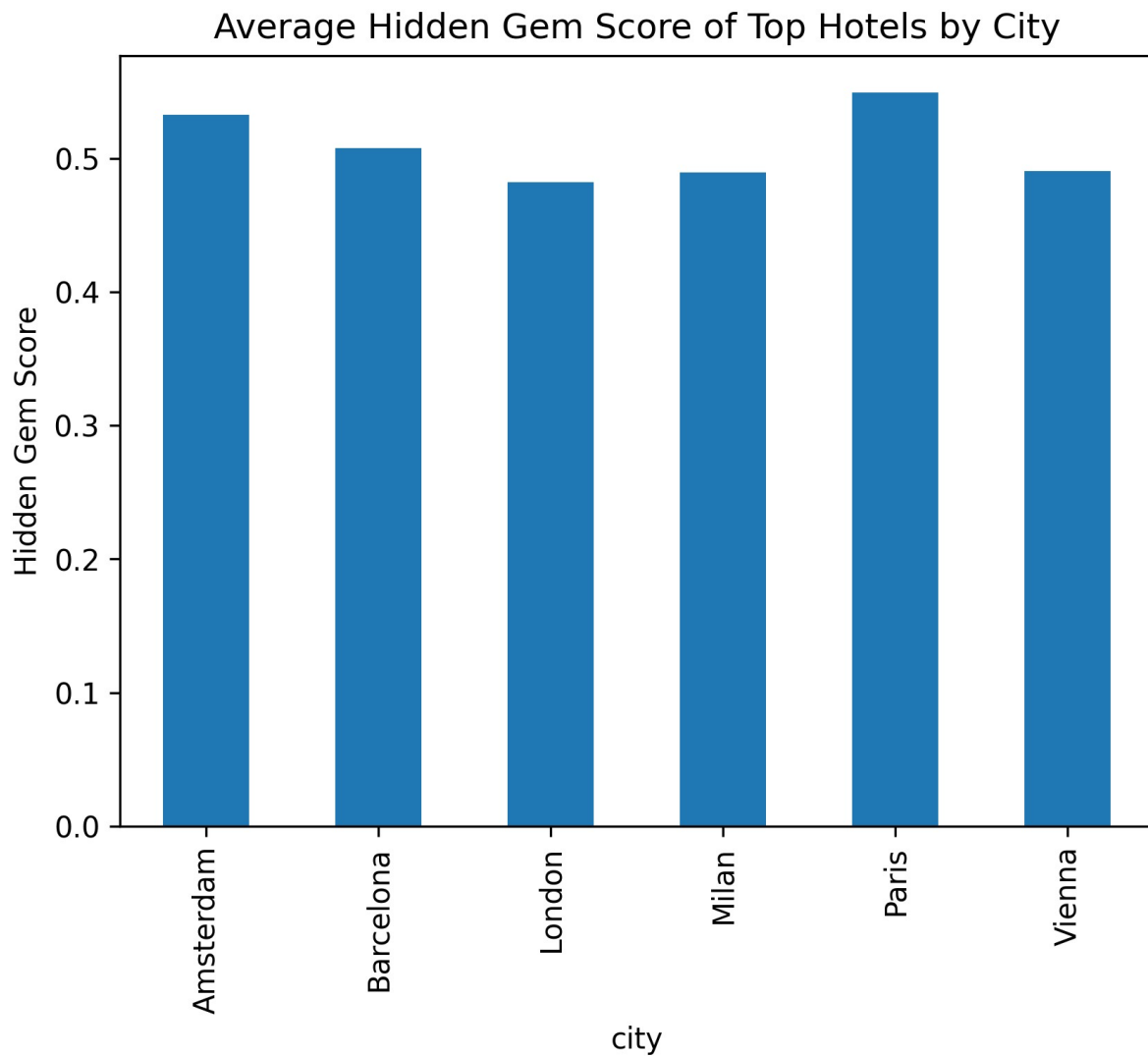


Figure 3. Average hidden-gem score of top hotels by city.

Takeaway

Top hidden-gem opportunities vary by city, with Paris and Amsterdam showing the strongest average top-hotel scores in this sample.

Impact of User Preferences on Paris Hotel Recommendations

Rank	Quiet Traveler	Lively Traveler
1	H tel Le Walt	Hotel Le Mareuil
2	Mercure Paris Gobelins Place d'Italie	Splendide Royal Paris
3	Hotel Tour d'Auvergne Opera	Hotel Whistler
4	H tel Baume	H tel Le Walt
5	Hotel Villa Lafayette Paris IX	Shangri La Hotel Paris
6	H tel Champs Elyses Plaza	Lyric H tel Paris
7	Mandarin Oriental Paris	H tel Saint Marc
8	H tel Amastan Paris	Hotel Villa Lafayette Paris IX
9	Victoire Germain	Mandarin Oriental Paris
10	H tel Arvor Saint Georges	H tel Monna Lisa Champs Elyses

7 of the top 10 recommendations changed when traveler preferences changed.

Figure 4. Impact of user preferences on Paris recommendations.

Takeaway

Traveler preferences changed 7 of the top 10 Paris recommendations, showing that personalization meaningfully affects rankings.

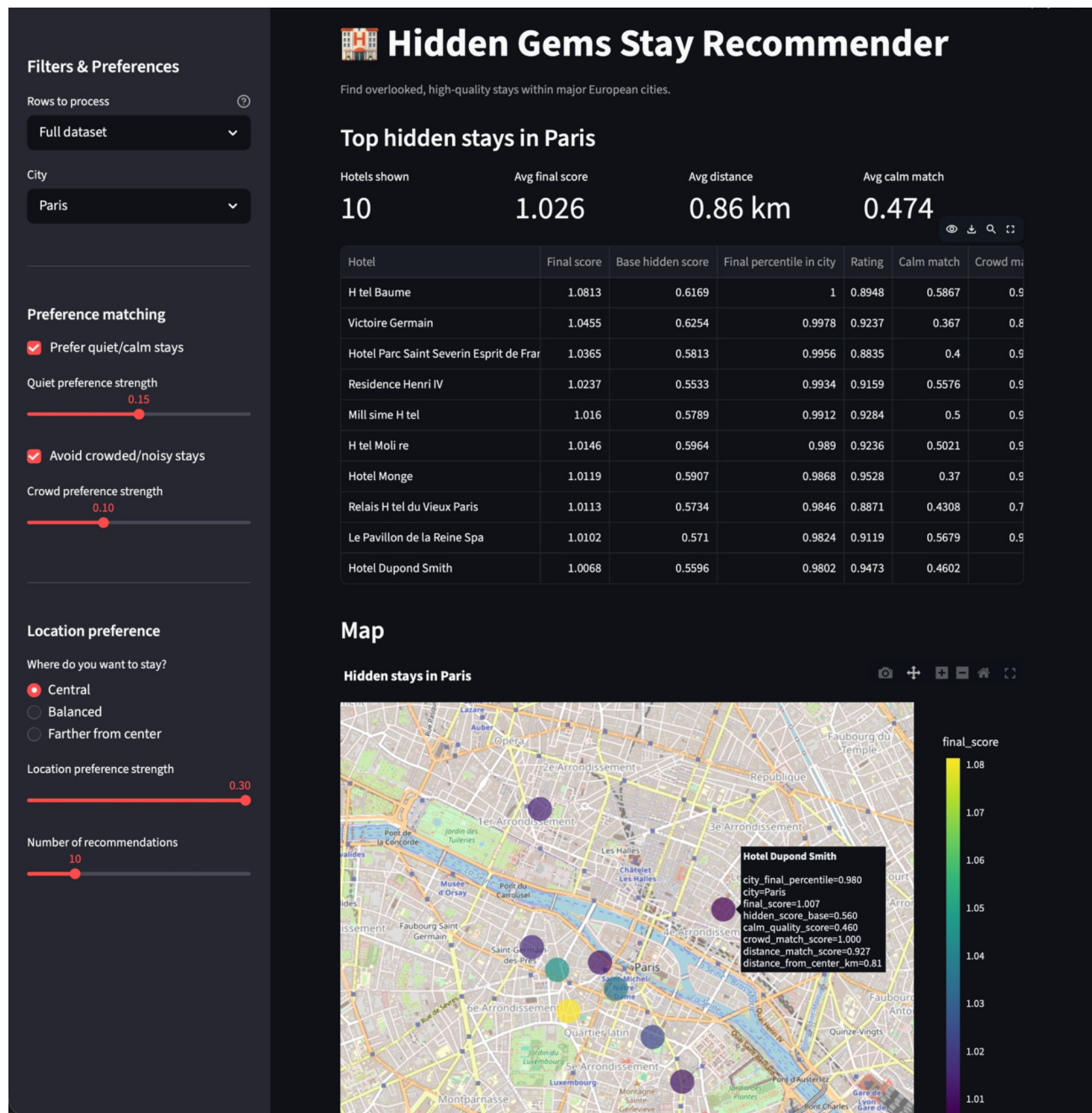


Figure 5. Streamlit application interface.

Takeaway

The app demonstrates the completed end-to-end deliverable: users can select preferences, view rankings, inspect map results, and review explainability metrics.

FINDINGS

Findings

- A relatively small group of hotels score higher than the main distribution, making hidden-gem ranking useful for discovery.
- Popularity is an imperfect proxy for quality; normalized review volume does not fully explain hidden-gem score.
- City-level differences matter, so rankings should be evaluated within each city rather than globally.
- Preference controls have a material effect on results, supporting the decision to separate baseline quality from traveler-specific matching.
- Explainability features make recommendations easier to interpret and debug.

RECOMMENDATIONS

Recommendations

1. Use the recommender as a discovery layer for travelers who want high-quality hotels beyond the most popular listings.
2. Present users with preference controls for quietness, crowd tolerance, and location style before displaying final rankings.
3. Show explainability metrics alongside each recommendation so users can understand whether a hotel matched because of calm quality, crowd profile, or location fit.

LIMITATIONS

Limitations

- Report figures are based on a 100,000-review development sample, not a full rerun of every notebook figure on all reviews.
- TextBlob is lightweight and interpretable but less context-aware than modern transformer sentiment models.
- Aspect keyword matching can miss nuanced language, sarcasm, multilingual phrasing, or implied sentiment.
- The project uses historical review data and does not include current prices, availability, renovation status, or recent ownership changes.
- The scoring weights are expert-defined rather than learned from observed booking or satisfaction outcomes.

FUTURE WORK

Future Work

- Replace or supplement TextBlob with transformer-based sentiment models for more context-aware aspect scoring.
- Incorporate current price, seasonality, and availability to improve real-world booking usefulness.
- Add attraction, transit, neighborhood, and safety datasets to strengthen geographic personalization.

- Learn preference weights from user feedback once real usage data is available.
- Evaluate recommendations with user studies or A/B testing rather than relying only on engineered scoring logic.

CONCLUSION

Conclusion

The Hidden Gems Hotels Recommender shows how review text, hotel metadata, geographic features, and user preferences can be combined into an interpretable content-based recommendation system. The project demonstrates that hotel discovery can be improved by looking beyond popularity and by allowing travelers to define what a good stay means for them.

Final Takeaway

Hidden-gem discovery works best when baseline quality signals are combined with preference-aware ranking and clear explanations.